

HEIDRUN, Norway

WMO: 011025

Lat: 65.333N

Lon: 7.317E

Elev: 80

StdP: 100.37

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.9	-1.9	-9.9	1.6	-0.7	-8.8	1.8	-0.4	29.8	5.3	27.7	5.3	11.3	100	1.123

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.4	17.8	14.9	16.2	14.1	15.2	13.3	15.3	17.1	14.5	16.1	13.8	15.3	8.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.2	10.2	16.0	13.8	9.9	15.5	13.0	9.4	14.5	43.1	17.0	40.9	15.7	38.6	15.3	17.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.2	23.3	21.0	DB	-4.4	19.9	1.4	1.9	-5.5	21.2	-6.3	22.3	-7.1	23.4	-8.1	24.7	(4)
(5)			WB	-5.7	16.3	1.2	1.2	-6.5	17.1	-7.2	17.8	-7.9	18.5	-8.8	19.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	3.8	3.7	3.7	5.4	7.3	9.4	12.2	12.7	11.0	8.0	5.6	4.5
	DBStd	4.01	2.53	2.54	2.52	2.47	2.46	2.17	2.32	1.90	1.97	2.37	2.60	2.53
	HDD10.0	1217	193	177	195	138	89	36	3	1	12	68	133	170
	HDD18.3	4027	452	411	453	387	342	267	190	174	220	320	383	428
	CDD10.0	233	0	0	0	1	6	19	72	86	42	7	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	10.5	12.7	12.1	11.7	10.1	8.8	8.3	8.2	8.6	10.3	11.3	11.9	12.8
Precipitation	PrecAvg	1022	131	99	82	56	39	48	62	83	97	106	101	118
	PrecMax	1236	266	175	143	110	65	100	90	128	146	190	162	201
	PrecMin	647	60	28	25	25	20	19	27	42	46	39	20	50
	PrecStd	146	49	41	33	21	11	17	16	26	30	43	37	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	8.2	9.1	12.8	16.0	17.8	20.0	19.0	16.1	13.8	11.0	9.2
		MCWB	7.8	6.9	6.8	8.8	11.6	13.3	15.8	15.3	13.5	11.5	9.6	8.1
	2%	DB	8.1	8.1	8.1	10.8	13.2	15.0	18.1	17.2	15.0	12.2	10.1	9.0
		MCWB	7.0	6.8	6.6	8.0	10.2	12.3	15.3	14.8	13.0	10.4	9.0	8.1
	5%	DB	7.9	7.2	7.8	9.1	11.9	13.8	16.8	16.2	14.1	11.8	9.2	8.2
		MCWB	6.7	5.9	6.4	7.5	9.4	11.4	14.6	14.3	12.5	10.2	8.2	7.1
	10%	DB	7.1	7.0	7.1	8.1	10.2	12.2	15.2	15.2	13.2	11.0	9.0	7.9
		MCWB	5.8	5.7	5.9	7.0	8.6	10.5	13.4	13.6	11.9	9.6	8.0	6.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	7.7	7.7	9.3	11.6	13.9	16.6	16.2	13.9	11.8	10.2	8.9
		MCDB	8.3	7.9	8.2	11.4	14.9	16.6	18.8	18.2	15.5	13.0	10.5	8.9
	2%	WB	7.3	7.0	7.0	8.3	10.6	12.7	15.6	15.3	13.3	11.1	9.2	8.1
		MCDB	7.7	7.5	7.5	9.6	12.9	14.7	17.6	16.9	14.5	12.2	9.6	8.4
	5%	WB	6.6	6.3	6.5	7.7	9.7	11.7	14.7	14.7	13.0	10.5	8.6	7.4
		MCDB	7.3	7.1	7.2	8.6	11.3	13.2	16.5	16.1	14.1	11.5	9.1	8.0
	10%	WB	5.8	5.6	5.8	7.2	8.9	11.0	13.8	14.0	12.5	9.9	7.8	6.6
		MCDB	6.9	6.7	6.8	8.0	10.1	12.4	15.5	15.3	13.6	10.7	8.7	7.5
Mean Daily Temperature Range	5% DB	MDBR	3.0	3.0	3.0	2.8	2.8	2.4	2.5	2.4	2.4	2.3	2.9	3.0
		MCDBR	3.1	3.0	3.2	3.7	4.5	4.4	4.9	3.8	3.2	2.6	2.8	3.3
		MCWBR	3.6	3.3	3.4	2.8	2.8	2.9	2.9	2.8	2.5	2.6	3.0	3.7
	5% WB	MCDBR	3.1	3.0	3.2	3.2	4.1	4.0	4.6	3.5	3.0	2.5	2.8	3.2
		MCWBR	3.9	3.7	3.8	2.8	2.9	2.9	3.0	2.9	2.7	2.8	3.4	4.1
Clear-Sky Solar Irradiance	taub	0.374	0.325	0.341	0.361	0.350	0.349	0.357	0.351	0.340	0.320	0.357	0.372	
	taud	2.392	2.185	2.306	2.347	2.456	2.497	2.496	2.507	2.488	2.297	2.426	2.362	
	Ebn at Noon	205	534	726	805	857	866	846	815	737	542	210	49	
	Edh at Noon	24	57	79	96	94	93	90	81	67	50	21	11	
All-Sky Solar Radiation	RadAvg	0.08	0.44	1.56	3.26	4.80	5.01	4.50	3.44	2.06	0.78	0.15	0.02	
	RadStd	0.01	0.06	0.15	0.30	0.25	0.56	0.50	0.32	0.24	0.12	0.02	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page